

clean image



adversarial image



perturbation ($\times 16$, centered)



Gemma-3-12B · GQA

correct answer

Q: What is the island made of?

A (clean = adv): "Wood"

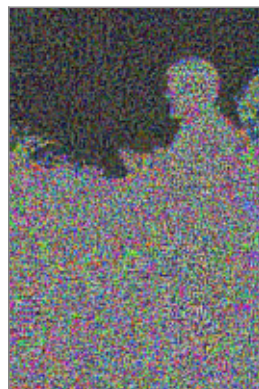
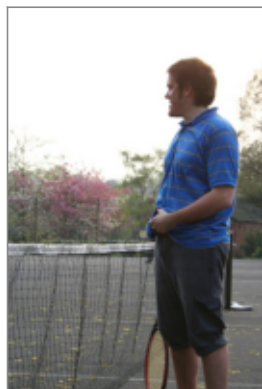
readout: FULL gate: 10% clean-FA

score 0.926 $\rightarrow$ 0.850 ($\tau=0.872$)

gate **ACCEPT** $\rightarrow$ **REJECT**

correct answer now gated out

$\ell_\infty=8.0/255$ · byte-identical answer



Gemma-3-12B · GQA

wrong answer

Q: Does the sky look cloudy?

A (clean = adv): "No"

readout: FULL gate: 10% clean-FA

score 0.802 $\rightarrow$ 0.874 ($\tau=0.872$)

gate **REJECT** $\rightarrow$ **ACCEPT**

wrong answer now admitted

$\ell_\infty=8.0/255$ · byte-identical answer



InternVL3.5-2B · POPE

correct answer

Q: Is there a traffic light in the image?

A (clean = adv): "No"

readout: FULL gate: 10% clean-FA

score 0.996 $\rightarrow$ 0.331 ($\tau=0.995$)

gate **ACCEPT** $\rightarrow$ **REJECT**

correct answer now gated out

$\ell_\infty=8.0/255$ · byte-identical answer



InternVL3.5-2B · POPE

wrong answer

Q: Is there a backpack in the image?

A (clean = adv): "No"

readout: FULL gate: 10% clean-FA

score 0.632 $\rightarrow$ 0.996 ($\tau=0.995$)

gate **REJECT** $\rightarrow$ **ACCEPT**

wrong answer now admitted

$\ell_\infty=8.0/255$ · byte-identical answer